

MATH70102 UNSUPERVISED LEARNING

LIVE SESSION — WEEK 4

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IMPERIAL

Review of this week's content

Computational examples

Problem sheet questions

Independent component analysis

Overview

- **Independent component analysis** (ICA) seeks to extract signals observed through **linear mixtures**
- **Key assumption 1:** The signals are mutually **independent**
- **Key assumption 2:** The signals are **non-Gaussian** (not normally distributed)
- Like PCA, ICA is a **data rotation technique**, but with different objective
- Instead of variance, **maximise independence and non-Gaussianity**

Independent component analysis (cont'd)

Cocktail-party problem

- **Two speakers**, with speech signals $S_1(t)$ and $S_2(t)$, in the same room
- **Two microphones** placed in the room with recorded signals $X_1(t)$ and $X_2(t)$ (e.g., **left and right channels**)
- Ignoring acoustics (and other complications), $X_1(t)$ and $X_2(t)$ are **linear mixtures**:

$$X_1(t) = a_{1,1}S_1(t) + a_{1,2}S_2(t)$$

$$X_2(t) = a_{2,1}S_1(t) + a_{2,2}S_2(t)$$

- **Problem**: extract $S_1(t)$ and $S_2(t)$ from $X_1(t)$ and $X_2(t)$

Independent component analysis (cont'd)

ICA model

- Observed d -dimensional data are generated by a random vector $\mathbf{X}$ such that

$$\mathbf{X} = \mathbf{A}\mathbf{S},$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ and $\mathbf{S}$ is a d -dimensional random vector with **independent components** such that $\mathbb{E}[\mathbf{S}] = \mathbf{0}$ and $\mathbb{E}[\mathbf{S}\mathbf{S}^T] = \mathbf{I}_d$

- Standard to assume $\mathbf{X}$ is **whitened** (cf. white noise):

$$\mathbb{E}[\mathbf{X}] = \mathbf{0}, \quad \mathbb{E}[\mathbf{X}\mathbf{X}^T] = \mathbf{I}_d$$

- Then $\mathbf{A}$ must be **orthogonal**

Independent component analysis (cont'd)

Measuring independence

- Objective is to find orthogonal $A \in \mathbb{R}^{d \times d}$ such that the components of

$$\mathbf{Y} := A^T \mathbf{X}$$

are as **independent as possible**

- Differential entropy** of $\mathbf{Y}$ with pdf $g_{\mathbf{Y}}$:

$$H(\mathbf{Y}) := \mathbb{E}[-\log g_{\mathbf{Y}}(\mathbf{Y})] = - \int_{\mathbb{R}^d} g_{\mathbf{Y}}(\mathbf{y}) \log g_{\mathbf{Y}}(\mathbf{y}) d\mathbf{y}$$

Independent component analysis (cont'd)

Measuring independence (cont'd)

- **Mutual information** between the components $Y_1, \dots, Y_d$ of $\mathbf{Y}$:

$$I(\mathbf{Y}) := \sum_{i=1}^d H(Y_i) - H(\mathbf{Y}) \geq 0$$

- We have $I(\mathbf{Y}) = 0$ when $Y_1, \dots, Y_d$ mutually **independent**
- In the current case $\mathbf{Y} := A^T \mathbf{X}$ with orthogonal A ,

$$I(\mathbf{Y}) := \sum_{i=1}^d H(Y_i) - H(\mathbf{X})$$

Independent component analysis (cont'd)

Maximising independence and minimising Gaussianity

- To **maximise independence**, we minimise

$$\sum_{i=1}^d H(Y_i)$$

with respect to A

- Among all random variables Y with zero mean and variance σ^2 , $H(Y)$ maximised by $Y \sim N(0, \sigma^2)$
- Hence, when minimising $\sum_{i=1}^d H(Y_i)$, we are also **minimising the Gaussianity** of $Y_1, \dots, Y_d$

Independent component analysis (cont'd)

Practical solution

- $H(Y_i)$ is **hard to work with** in practice — it depends on the pdf of Y_i , which depends on A (which we are optimising over)
- Replace $H(Y_i)$ with **negentropy** (which is maximised)

$$J(Y_i) := H(Z_i) - H(Y_i), \quad \text{where } Z_i \sim N(0, \text{Var}[Y_i])$$

- **Approximate:**

$$J(Y_i) \approx (\mathbb{E}[G(Y_i)] - \mathbb{E}[G(Z_i)])^2$$

using, e.g., $G(x) := \frac{1}{a} \log \cosh(ax)$ with $a \in [1, 2]$

Independent component analysis (cont'd)

Practical solution (cont'd)

- Optimisation is then carried out using **Newton's method**

Caveats of ICA

- Unable to recover **variances** of the components (unlike PCA)
- Components are in **arbitrary order** — not in the order of importance (unlike PCA)

Multidimensional scaling

Overview

- Multidimensional scaling (MDS) seeks to **map** samples in a high-dimensional space into a lower dimensional one by **preserving their geometry** as much as possible
- Geometry can be measured via:
 - **similarities** (scalar products)
 - **dissimilarities** (distances)
- MDS does **not** require the actual samples — just their similarities or dissimilarities

Multidimensional scaling (cont'd)

MDS setting

- Map $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ to $\mathbf{z}_1, \dots, \mathbf{z}_n \in \mathbb{R}^q$ ($q \leq n$)
- Try to preserve **distances** (**dissimilarities**):

$$\underbrace{\|\mathbf{x}_i - \mathbf{x}_j\|}_{=:d_{ij}} \approx \|\mathbf{z}_i - \mathbf{z}_j\|$$

- ...or centred **scalar products** (**similarities**):

$$\underbrace{(\mathbf{x}_i - \bar{\mathbf{x}})^T (\mathbf{x}_j - \bar{\mathbf{x}})}_{=:s_{ij}} \approx (\mathbf{z}_i - \bar{\mathbf{z}})^T (\mathbf{z}_j - \bar{\mathbf{z}})$$

Multidimensional scaling (cont'd)

MDS objective functions (to minimise)

1. Classical scaling (strain):

$$S_C(\mathbf{z}_1, \dots, \mathbf{z}_n) := \sum_{i,j} (s_{i,j} - (\mathbf{z}_i - \bar{\mathbf{z}})^T (\mathbf{z}_j - \bar{\mathbf{z}}))^2$$

2. Least squares scaling (stress):

$$S_M(\mathbf{z}_1, \dots, \mathbf{z}_n) := \sum_{i \neq j} (d_{i,j} - \|\mathbf{z}_i - \mathbf{z}_j\|)^2$$

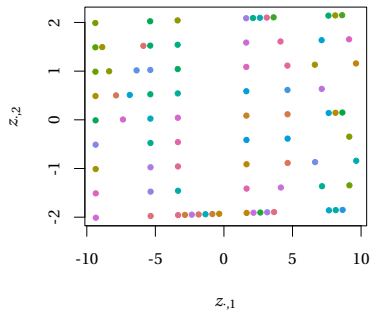
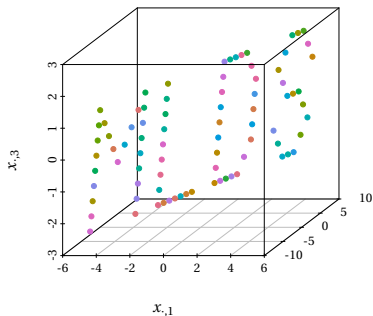
Multidimensional scaling (cont'd)

MDS objective functions (to minimise; cont'd)

3. Sammon mapping:

$$S_{Sm}(\mathbf{z}_1, \dots, \mathbf{z}_n) := \sum_{i \neq j} \frac{(d_{i,j} - \|\mathbf{z}_i - \mathbf{z}_j\|)^2}{d_{i,j}}.$$

Multidimensional scaling (cont'd)

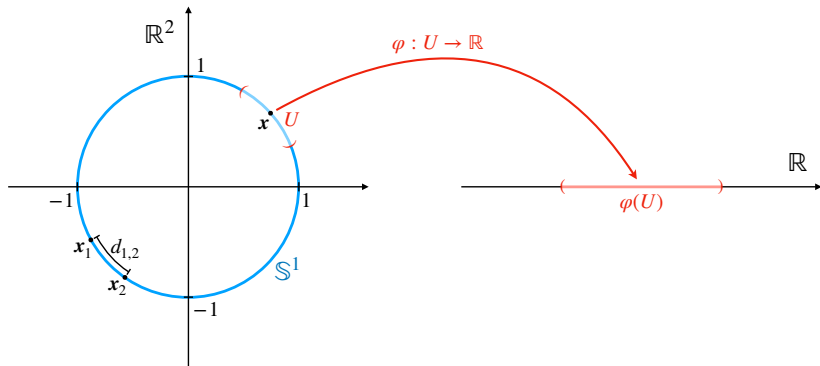


Non-linear dimensionality reduction

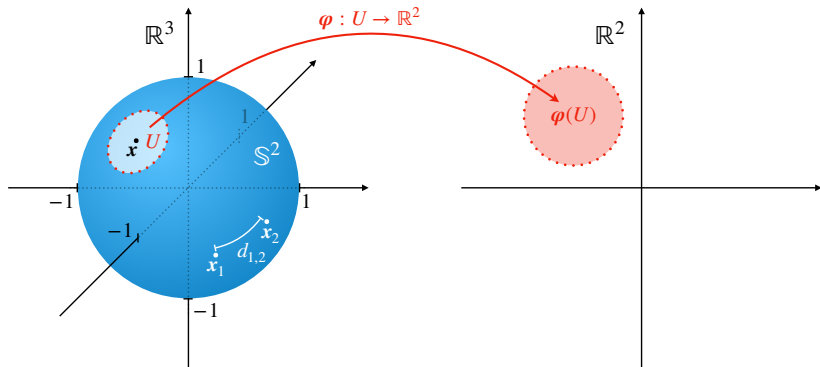
Overview

- Objective: reduce dimensionality of samples in $\mathbb{R}^d$ while preserving **local geometry**
- Samples on a **manifold** (possibly **non-linear** subspace of $\mathbb{R}^d$) — the **manifold hypothesis**
- A **neighbourhood** of a q -manifold in $\mathbb{R}^d$ looks a like neighbourhood of $\mathbb{R}^q$
- Local geometry $\leftrightarrow$ **geometry on the manifold**
- Difficulty: typically we do **not** know at the outset what the manifold looks like and **how to measure distances on it**

Non-linear dimensionality reduction (cont'd)



Non-linear dimensionality reduction (cont'd)



Non-linear dimensionality reduction (cont'd)

Methods

- Sammon mapping
- Isometric feature mapping (ISOMAP)
- Local linear embedding
- Local MDS
- t -Distributed stochastic neighbour embedding (t -SNE)

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Computational examples

Problem sheet questions

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Problem 1

Question

Load the data set `cpus.csv` provided in the zip file `cpus.zip`. The data set contains some **characteristics of 209 historical computer CPUs**. (The accompanying file `cpus.info.txt` provides explanations of the variables.)

Discard first the variable `name`, which contains the name of each computer. Then select the performance measure variables `perf` and `estperf` for a data set X and the remaining, descriptive variables for a data set Y .

Apply **canonical correlation analysis** to the data sets X and Y and try to interpret the results.

Problem 1

Solution

→ **Jupyter Notebook**

Problem 2

Question

Show that the mutual information $I(\mathbf{Y})$ between the components of a random vector $\mathbf{Y}$ is zero when the components are mutually independent.

Problem 2

Solution

By definition,

$$I(\mathbf{Y}) := \sum_{i=1}^d H(Y_i) - H(\mathbf{Y}),$$

where

$$H(\mathbf{Y}) = \mathbb{E}[-\log g_{\mathbf{Y}}(\mathbf{Y})]$$

When the components of $\mathbf{Y} = (Y_1, \dots, Y_d)$ are **mutually independent**,

$$g_{\mathbf{Y}}(y_1, \dots, y_d) = \prod_{i=1}^d g_{Y_i}(y_i)$$

Problem 2

Solution (cont'd)

Hence,

$$\begin{aligned}
 H(\mathbf{Y}) &= \mathbb{E}[-\log g_{\mathbf{Y}}(Y_1, \dots, Y_d)] \\
 &= \mathbb{E}\left[-\log \prod_{i=1}^d g_{Y_i}(Y_i)\right] = \mathbb{E}\left[-\sum_{i=1}^d \log g_{Y_i}(Y_i)\right] \\
 &= \sum_{i=1}^d \mathbb{E}\left[-\log g_{Y_i}(Y_i)\right] = \sum_{i=1}^d H(Y_i),
 \end{aligned}$$

whereby

$$I(\mathbf{Y}) = \sum_{i=1}^d H(Y_i) - \sum_{i=1}^d H(Y_i) = 0$$

Problem 3

Question

Consider a data set

$$\mathbf{x}_1 = (-1, 10), \quad \mathbf{x}_2 = (0, 0), \quad \mathbf{x}_3 = (1, 10),$$

in $\mathbb{R}^2$. These samples lie all on a **1-manifold**

$$M = \{(x, y) \in \mathbb{R}^2 : y = 10x^2\}.$$

(a) Find the **Euclidean distances** between $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ in $\mathbb{R}^2$.

Problem 3

Question (cont'd)

- (b) Find the **geodesic distances** (shortest paths) between $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ on M . To this end, it is useful to recall that the length of the graph of a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ between a and b (where $a < b$) is

$$\int_a^b \sqrt{1 + f'(x)^2} dx.$$

- (c) Map the samples $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ to z_1 , z_2 and z_3 on the real line $\mathbb{R}$ while trying to **preserve their Euclidean distances** as much as possible. *Hint:* Fix $z_1 = 0$ and optimise over z_2 and z_3 numerically or analytically.
- (d) Then do the same while trying to **preserve their geodesic distances**. Compare the results with those in (c).

Problem 3

Solution

→ **Jupyter Notebook**